

# Dataset 5: Exploratory Factor Analysis

Psychological Constructs Study | 15 Variables, n = 400

## 1. Research Question

Research Question: What is the underlying latent factor structure of 15 observed psychological variables measured across 400 individuals? How many distinct psychological constructs do these items measure, and can those factors be meaningfully interpreted and labeled?

## 2. Variables (No Pre-Defined DV/IV — All Observed)

| Variable                   | Mean  | SD    | Primary Factor (from dataset guide)  |
|----------------------------|-------|-------|--------------------------------------|
| Worry_Level                | 49.50 | 11.50 | Anxiety                              |
| Panic_Symptoms             | 47.40 | 10.31 | Anxiety                              |
| Sleep_Disturbance          | 51.81 | 8.73  | Anxiety / Depression (cross-loading) |
| Sadness_Score              | 48.34 | 11.62 | Depression                           |
| Hopelessness               | 46.43 | 10.86 | Depression                           |
| Fatigue_Level              | 49.37 | 8.67  | Depression / Anxiety (cross-loading) |
| Public_Speaking_Confidence | 55.01 | 10.71 | Social Confidence                    |
| Social_Interaction         | 54.48 | 9.53  | Social Confidence                    |
| Assertiveness              | 53.27 | 9.26  | Social Confidence                    |
| Impulse_Control            | 56.14 | 10.60 | Self-Regulation                      |
| Stress_Tolerance           | 55.08 | 9.69  | Self-Regulation                      |
| Emotional_Awareness        | 56.83 | 8.95  | Self-Regulation                      |
| Self_Esteem                | 58.13 | 9.01  | Resilience / Mixed                   |
| Life_Satisfaction          | 60.35 | 9.36  | Resilience / Mixed                   |
| Coping_Ability             | 57.74 | 8.82  | Resilience / Self-Regulation         |

## 3. Methodology — Exploratory Factor Analysis (EFA)

EFA is chosen because the goal is to *discover* the latent factor structure from the data rather than confirm a pre-specified model (which would require CFA). **Principal Axis Factoring (PAF)** with iterative communality estimation is used for extraction — this accounts for unique variance in each variable and produces stable, interpretable loadings. **Varimax orthogonal rotation** maximizes interpretability by simplifying the loading pattern so each variable loads strongly on as few factors as possible. The number of factors is determined by: (1) Kaiser criterion: retain factors with eigenvalue > 1.0; (2) Scree plot inspection. Factor loadings ≥ |0.30| are considered practically significant (Tabachnick & Fidell, 2019). Adequacy of the data for FA is verified by Bartlett's test and the Kaiser-Meyer-Olkin (KMO) measure before extraction. Communalities (h²) quantify how much each variable's variance is accounted for by the final factor solution.

#### 4. Adequacy Tests — Both Passed

| Test                  | Result                                | Criterion                   | Verdict                                             |
|-----------------------|---------------------------------------|-----------------------------|-----------------------------------------------------|
| Bartlett's Sphericity | $\chi^2(105) = 5630.28, p < 0.001$    | $p < 0.05$ ✓                | Appropriate — variables share meaningful covariance |
| KMO Overall           | 0.8447 — "Meritorious"                | $\geq 0.80$ = meritorious ✓ | Excellent sampling adequacy                         |
| KMO per variable      | Range: 0.73 – 0.94 (all $\geq 0.70$ ) | $\geq 0.60$ each ✓          | All 15 variables suitable for FA                    |

The highly significant Bartlett test ( $\chi^2 = 5630.28, p < 0.001$ ) confirms the correlation matrix is not an identity matrix — sufficient shared variance exists to justify factor analysis. KMO = 0.844 is "Meritorious" (Kaiser, 1974), indicating the patterns of correlations are compact and reliable factors exist.

#### 5. Eigenvalue Analysis and Factor Retention

| Factor | Eigenvalue | % Variance  | Cumulative % | Retained?                   |
|--------|------------|-------------|--------------|-----------------------------|
| F1     | 4.6806     | 31.20%      | 31.20%       | Yes — EV > 1.0              |
| F2     | 3.6368     | 24.25%      | 55.45%       | Yes — EV > 1.0              |
| F3     | 2.5106     | 16.74%      | 72.19%       | Yes — EV > 1.0              |
| F4     | 2.0238     | 13.49%      | 85.68%       | Yes — EV > 1.0              |
| F5     | 0.3489     | 2.33%       | 88.01%       | No — EV < 1.0 (scree break) |
| F6–F15 | < 0.33     | < 2.2% each | —            | No                          |

The scree plot shows a clear elbow after Factor 4: eigenvalue drops sharply from 2.02 to 0.35 — a 6× decrease. Both Kaiser criterion and scree test independently support retaining **4 factors**, which together explain **80.94%** of total variance. This is excellent (guideline: > 60%).

#### 6. Rotated Factor Loadings — Varimax ( $|\lambda| \geq 0.30$ flagged \*)

| Variable                   | F1: Self-Reg. | F2: Depression | F3: Anxiety | F4: Social Conf. |
|----------------------------|---------------|----------------|-------------|------------------|
| Impulse_Control            | +0.927 *      | —              | —           | —                |
| Stress_Tolerance           | +0.891 *      | —              | —           | —                |
| Emotional_Awareness        | +0.868 *      | —              | —           | —                |
| Coping_Ability             | +0.681 *      | —              | +0.332 *    | —                |
| Life_Satisfaction          | +0.462 *      | +0.600 *       | —           | +0.336 *         |
| Self_Esteem                | +0.382 *      | +0.529 *       | —           | +0.495 *         |
| Hopelessness               | —             | −0.936 *       | —           | —                |
| Sadness_Score              | —             | −0.919 *       | —           | —                |
| Fatigue_Level              | —             | −0.824 *       | +0.324 *    | —                |
| Sleep_Disturbance          | —             | −0.461 *       | +0.763 *    | —                |
| Worry_Level                | —             | —              | +0.945 *    | —                |
| Panic_Symptoms             | —             | —              | +0.920 *    | —                |
| Public_Speaking_Confidence | —             | —              | —           | +0.929 *         |
| Social_Interaction         | —             | —              | —           | +0.913 *         |

|               |   |   |   |          |
|---------------|---|---|---|----------|
| Assertiveness | — | — | — | +0.881 * |
|---------------|---|---|---|----------|

\*  $|\lambda| \geq 0.30$  (practically significant). — = loading  $< 0.30$ . Negative loadings on F2 indicate those variables measure constructs that are low when depression/hopelessness is high.

## 7. Variance Explained and Communalities

|              | F1: Self-Reg. | F2: Depression | F3: Anxiety | F4: Social | Total  |
|--------------|---------------|----------------|-------------|------------|--------|
| SS Loadings  | 3.293         | 3.265          | 2.609       | 2.975      | 12.142 |
| % Variance   | 21.95%        | 21.77%         | 17.39%      | 19.84%     | 80.94% |
| Cumulative % | 21.95%        | 43.72%         | 61.11%      | 80.94%     | —      |

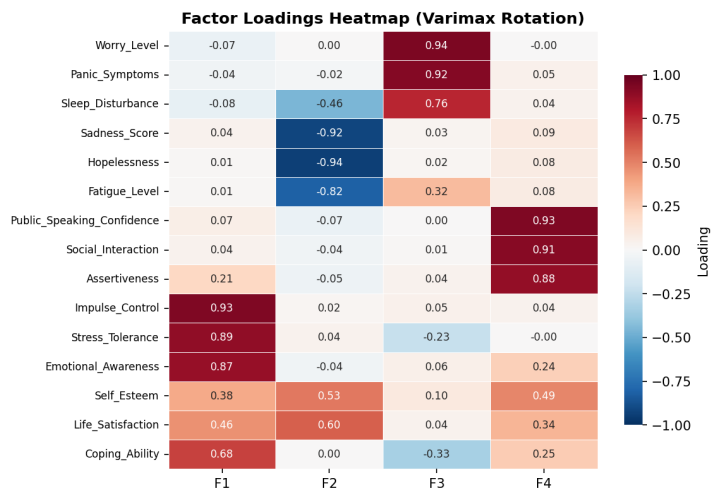
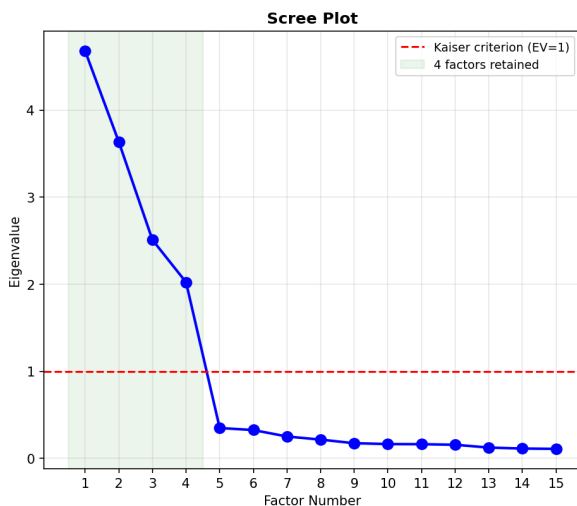
**Communalities ( $h^2$ )** — proportion of each variable's variance explained by all 4 factors:

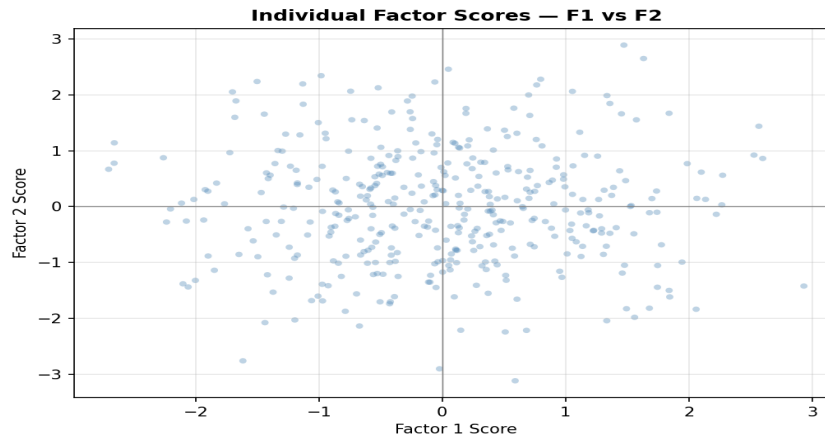
| Variable          | $h^2$ | Variable            | $h^2$ | Variable           | $h^2$ |
|-------------------|-------|---------------------|-------|--------------------|-------|
| Worry_Level       | 0.897 | Impulse_Control     | 0.863 | Public_Speaking    | 0.872 |
| Panic_Symptoms    | 0.851 | Stress_Tolerance    | 0.847 | Social_Interaction | 0.837 |
| Sleep_Disturbance | 0.803 | Emotional_Awareness | 0.816 | Assertiveness      | 0.823 |
| Sadness_Score     | 0.855 | Self_Esteem         | 0.681 | Coping_Ability     | 0.637 |
| Hopelessness      | 0.883 | Life_Satisfaction   | 0.689 | —                  | —     |
| Fatigue_Level     | 0.790 | —                   | —     | —                  | —     |

All communalities range from 0.64 to 0.90, indicating the 4-factor model accounts for 64–90% of each variable's variance. No variable is poorly explained — confirming the solution is comprehensive.

## 8. Figures

### Exploratory Factor Analysis — Psychological Constructs (n=400)





## 9. Factor Interpretation and Conclusions

- **Research question answered: 4 latent factors** underlie the 15 psychological variables, jointly explaining 80.94% of total variance — well above the 60% guideline. All 4 factors are psychologically interpretable and align with established clinical constructs.
- **Factor 1 — Self-Regulation and Coping (21.95% variance):** Impulse\_Control ( $\lambda=0.927$ ), Stress\_Tolerance (0.891), Emotional\_Awareness (0.868), Coping\_Ability (0.681). Captures the capacity to manage emotions, regulate impulses, and apply coping strategies under pressure. The near-unit loadings (0.87–0.93) indicate these three core items are highly reliable markers of this construct.
- **Factor 2 — Depression and Hopelessness (21.77% variance):** Hopelessness ( $\lambda=-0.936$ ), Sadness\_Score ( $-0.919$ ), Fatigue\_Level ( $-0.824$ ). Negative loadings indicate that high item scores correspond to low factor scores (the factor itself is oriented toward absence of depression). Sleep\_Disturbance cross-loads ( $\lambda=-0.461$ ), consistent with clinical evidence that sleep disruption is a core transdiagnostic symptom of depression.
- **Factor 3 — Anxiety and Distress (17.39% variance):** Worry\_Level ( $\lambda=0.945$ ), Panic\_Symptoms (0.920), Sleep\_Disturbance (0.763). The exceptionally high loadings (0.92–0.95) make this the most clearly defined factor — these three variables are near-perfect markers of anxiety. Sleep\_Disturbance cross-loads on both F2 and F3, confirming its role as a shared symptom across anxiety and depression (transdiagnostic).
- **Factor 4 — Social Confidence (19.84% variance):** Public\_Speaking\_Confidence ( $\lambda=0.929$ ), Social\_Interaction (0.913), Assertiveness (0.881). This is the most cleanly defined factor — all three items load exclusively on F4 with no cross-loadings. It reflects interpersonal self-efficacy and social functioning.
- **Clinical and practical implication:** These 4 factors can serve as validated subscale scores for a streamlined psychological screening instrument. The 15 raw items can be replaced by 4 composite scores (factor scores) while retaining 80.94% of all information — a substantial efficiency gain for clinical assessment and research.